SECP3213: BUSINESS INTELLIGENCE – SEMESTER 2, 2025/2026

Assignment 1: Understanding Business Intelligence & Decision-Making

"Think like a decision-maker. If you have data, how would you use it to solve the problem?"

1. Assignment Overview

This group assignment is designed to help you understand how **Business Intelligence (BI)** supports **data-driven decision-making** in real-world contexts. Your group is required to analyze a given problem scenario, identify decision-making levels, and explain how BI and basic BI architecture can be used to support better decisions.

2. Learning Outcome

At the end of this assignment, students will be able to:

Explain the concepts, role, and basic architecture of Business Intelligence in supporting decision-making. *(CLO1)*

3. Group and Assigned Topics

Each group must include **5 or 6 members.** Refer to the following table for your Group's topic:

Group	Topic
2	Improving Student Academic Performance
1	Increasing Sales in an Online Business (Shopee/TikTok Shop)
4	Improving Customer Ratings in Ride-Hailing (Grab)
3	Reducing Waiting Time in Cafeteria / Food Court
6	Improving Appointment Scheduling in Clinic / Hospital
5	Optimizing Classroom / Lab Usage in University
7	Improving Customer Satisfaction in a Retail Store

4. Assignment Requirements

The assignment consists of **THREE (3) parts**:

Part 1: Problem & Decision-Making Analysis (30%)

You group must:

1. Clearly describe the **problem scenario**
2. Identify and explain the **three levels of decision-making**:
    - Operational
    - Tactical
    - Strategic
3. Provide **one example decision for each level**

Part 2: Role of Business Intelligence (40%)

Your group should explain:

1. **How BI can help solve the problem**
2. At least **THREE (3) benefits of BI** in this context
3. At least **TWO (2) data-related challenges**, such as:
    - Data scattered across systems
    - Data inconsistency
    - Lack of access to data

Part 3: Basic BI Architecture (30%)

Your group should describe a **simple BI process (specific to your topic)**, including:

1. **Data Sources**
    - Where the data comes from
2. **Data Flow**
    - From operational systems to storage and analysis
3. **Output / Information**
    - What insights are generated for decision-making
4. Include **ONE diagram** illustrating the BI process

Example format: Data Source → Data Storage → Analysis → Decision

5. Deliverables

Your group should submit a **report:**

- Format: Times New Roman, 12pt, 1.5 spacing
- Length: **minimum 10 pages** (including 1 cover page & 1 page for references)
- Include any relevant figures and tables to support your content
- Include an appendix that informs:
    - The GenAI tool is/used : ChatGPT, Gemini, DeepSeek or other tools
    - The prompts used

6. Submission Details

- Submission method: Submit via eLearning – ONE (1) submission only per group
- Deadline: 8 April 2026, 11.59 pm
- File format: PDF – Important: Name the file ‘Asg1_GroupX.pdf’

7. Assessment Rubric (10 Marks)

Criteria	Marks (%)
Problem clarity & decision levels	3
Understanding of BI concepts & benefits	3
BI architecture explanation	2
Diagram clarity & relevance	1
Report structure & presentation	1
Total	**10**